Anton Roupassov-Ruiz

Honors Computing Science @ University of Alberta · Software Engineer Intern @ Microsoft

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Education

BSc **University of Alberta**, Computing Science (Honors) — Edmonton, AB, Canada — Sept 2023 – June 2027

- cGPA: 3.54 / 4.00
- Relevant coursework: Reinforcement Learning, Machine Learning, Search & Planning in AI, Program Synthesis, Heuristic Search & Artificial Life Research Seminar, Software Engineering Principles, Data Structures & Algorithms, Design & Analysis of Algorithms.

Experience

Microsoft, Software Engineer Intern — Redmond, WA, USA — May 2026 – present — 3 months

- Engineered a production-grade AI agent integrated with Model Context Protocol (MCP) tools for a new internal platform, letting users translate natural language into dynamically generated, debugged, and deployed enterprise agent workflows.
- Architected the backend to orchestrate model invocation via Azure AI Foundry and persist agent conversation histories in Azure Cosmos DB.
- Accelerated development cycles by leveraging the GitHub Copilot CLI to manage distributed agent fleets and autonomous subagents.

Microsoft, Explore (Software Engineering & Product Management) Intern — Redmond, WA, USA — May 2025 – Aug 2025 — 4 months

- Identified Bing ETL pipeline bottlenecks with PMs and engineers and implemented the optimal architectural approach.
- Built a full-stack tool (React/TypeScript frontend, Java backend) that cut ETL debugging latency from 15 minutes to seconds using Azure Kubernetes Service, Azure Functions, and Azure SignalR Service.
- Presented a live demo and quantified impact to stakeholders across Microsoft Bing, including CVPs and VPs.

University of Alberta — RLAI Lab (Dr. Vadim Bulitko), Undergraduate Researcher — Edmonton, AB, Canada — Sept 2025 – Dec 2025 — 4 months

- Developed the first fully specified, open-source reimplementation of the classic Ackley & Littman (1991) predator–prey evolutionary reinforcement learning A-Life testbed.
- Formulated explainable, differentiable decision lists as a programmatic alternative to black-box neural networks, evaluated across 4,000 independent trials with Kaplan–Meier survival analysis.
- Proved programmatic agents significantly outlasted neural networks (log-rank $p \sim$ 4.1e-72), surviving on average 201.69 steps longer (95% CI [180.14, 223.25]).

University of Alberta — Computational Psychology Lab (Dr. Jeremy Caplan), Undergraduate Researcher — Edmonton, AB, Canada — May 2024 – Aug 2024 — 4 months

- Authored a 4000+ word analysis of four computational associative memory models (grade 4.00/4.00) and implemented MATLAB simulations of neural-network, convolution, local-trace, and matrix models.

Projects

Prelude — Microsoft Global Intern Hackathon
Co-engineered a framework (team of 6) that automatically generates MCP tools for Microsoft Store apps by crawling the OS accessibility tree; demoed at the global hackathon science fair.

ERL — Evolutionary RL A-Life Testbed
Open-source reimplementation of the Ackley & Littman (1991) A-Life testbed with programmatic, explainable policies that significantly outlived neural networks. View on GitHub.

MentAI
Swift-based iOS MVP using the OpenAI API for LLM-assisted wellness insights; pitched to the Alberta Machine Intelligence Institute (Amii) and the UAlberta Health Sciences Startup Incubator.

EEG-Based Pomodoro Focus Timer
Python sensor-driven focus timer integrating EEG data streams and signal processing (BrainFlow) for the natHacks 2024 neurotech hackathon. View on GitHub.

Honors & Awards

- Dean's Honor Roll / First Class Standing — University of Alberta (2025, 2026)
- Jason Lang Scholarship — University of Alberta (2024, 2025)
- Alexander Rutherford Scholarship — Alberta Student Aid (2023)
- AP Scholar with Distinction — College Board (2023)

Leadership

- **Panelist**, "AI in Industry" — UAlberta Undergraduate AI Society (2025). Shared industry software-engineering experience at Microsoft.
- **Treasurer** — neurAlbertaTech, UAlberta Chapter (2024–2025). Oversaw finances and the annual budget for a student-led neurotechnology club.

Technical Skills

Languages: C#, TypeScript, Python, JavaScript, Java, C++, C, MATLAB, HTML/CSS, LaTeX

Tools & Cloud: React, Azure Kubernetes Service, Azure Functions, Azure SignalR, Azure AI Foundry, Cosmos DB, GitHub Copilot, Git, Flask

ML & Data: RL-Glue, Scikit-learn, Pandas, NumPy, Matplotlib

Languages

English
Native

Spanish
Native

Russian
Native

French
Intermediate